Ashish Molakalapalli

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EDUCATION

Master of Science in Computer Science — *Aug 2025 – May 2027*
Texas A&M University — *College Station, TX*

- GPA: 3.833 / 4.0 | Graduating **May 2027** | Coursework: Analysis of Algorithms, Operating Systems, Machine Learning, Artificial Intelligence, NLP Foundations & Tech, Information Storage & Retrieval
- Focus: Data Structures & Algorithms, System Design, Cloud Infrastructure, API Design, CI/CD and Release Engineering

Bachelor of Technology in Computer Science & Engineering — *Dec 2021 – Apr 2025*
IIIT Kottayam — *India*

- GPA: 9.34 / 10.0 | Selected Research Scholar, IoT Cloud Research Group – Top 5 of 150+ applicants

INDUSTRY EXPERIENCE

Software Engineering Intern – Security Platform — *May 2025 – Aug 2025*
ServiceNow — *San Diego, CA*

- Designed and shipped a modular 14-tool service layer in JavaScript orchestrating parallel database queries, REST callouts, and scheduled-job introspection across production security infrastructure with zero manual intervention.
- Built a schema-driven analysis pipeline with a 14-field JSON contract and confidence-based escalation logic; integrated AWS Bedrock as the inference backend, achieving consistent structured output across all instance types in production.
- Engineered a real-time alerting service delivering health summaries to Microsoft Teams via webhooks, replacing a manual multi-instance review process and cutting incident response latency from hours to minutes.
- Resolved cross-environment deployment conflicts during a production migration by auditing change sets, isolating conflicting records, and establishing a repeatable release protocol adopted by the team for subsequent releases.

Software Engineering Intern — *Sep 2024 – Jul 2025*
Shipsy — *Gurugram, India*

- Engineered event-driven backend automation integrating Slack, Gmail, and GitHub REST APIs to process 10K+ requests per day for DTDC; implemented request batching and async processing, reducing manual triage by 35% and cutting average API latency by 40%.
- Built a prototype knowledge graph service with structured entity linking and full-text search, improving internal documentation query response time by 40% under concurrent load.
- Designed and shipped an autonomous standup transcription pipeline with retry logic, idempotency guarantees, and Jira write-back; won Best Idea Award at Shipsy Hackathon 2025.

Software Developer Intern — *Jun 2024 – Aug 2024*
UST Global — *Kochi, India*

- Architected SmartChat as a modular microservices system; exposed NLP inference pipelines via production-grade REST endpoints with structured request validation, error handling, and response observability.
- Optimized request routing and introduced query-time caching, reducing average resolution latency by 30% and improving end-to-end accuracy by 25% under sustained load.

Software Intern — *Dec 2023 – Feb 2024*
Nav Tech — *Hyderabad, India*

- Implemented low-latency ingestion APIs for hardware data streams using async I/O and connection pooling, reducing average response time by 45%.
- Introduced health-check endpoints and alerting hooks integrated with the monitoring stack, sustaining 99.9% service uptime across the device fleet.

PROJECTS

Real-Time Job Ingestion and Deduplication Platform | *Python, GitHub Actions, Concurrent I/O, REST APIs, JavaScript*

- Built a concurrent data-ingestion service in Python polling 950+ external REST APIs across 6 platforms, sweeping 52K+ records per run with a bounded thread pool and per-source fault isolation.
- Deployed on an hourly CI/CD pipeline with tiered polling, byte-stable output to avoid churn commits, and write guards that refuse to overwrite good data when upstream sources degrade.

Scalable Task Management Platform | *React, Node.js, WebSockets, PostgreSQL, AWS EC2/RDS, Docker, JWT*

- Built a full-stack real-time collaboration application with a React front-end and Node.js back-end, WebSocket-driven live updates and optimistic UI; deployed containerized services on AWS EC2 with RDS PostgreSQL, sustaining 99.9% uptime under concurrent load.
- Implemented JWT authentication, role-based access control, and database indexing strategies to handle high-concurrency workloads without lock contention; load-tested to 500 concurrent users with sub-200ms P99 latency.

Financial Document Intelligence: Production RAG System | *RAG, GPT-4o, FAISS, BM25 Hybrid Search, Python, Flask, Docker, AWS EC2*

- Designed and deployed a production document search service over 100+ financial prospectuses using dense FAISS vector indexing with BM25 sparse re-ranking; improved retrieval precision by 32% over a TF-IDF baseline.
- Containerized the service with Docker and deployed on EC2; exposed low-latency REST endpoints with per-query source attribution, reducing analyst document retrieval time by 50% in production use.

E-Commerce Order Management Platform | *React, Node.js, MySQL, AWS ALB/RDS, Auto Scaling, Docker*

- Orchestrated checkout and order APIs with rate limiting, idempotency keys, and request validation; modeled the relational schema on RDS MySQL ensuring transactional integrity under concurrent load.
- Scaled horizontally via AWS ALB and auto-scaling groups; achieved consistent sub-300ms response times under 10x traffic spikes without manual intervention.

TECHNICAL SKILLS

Languages: Python, JavaScript (Node.js, React), C++, Java, SQL, Bash

Backend & Systems: REST APIs, Microservices, WebSockets, Async I/O, Event-Driven Architecture, Rate Limiting, Message Queuing, Concurrency

Databases: PostgreSQL, MySQL, Query Optimization, Indexing, Transactions, FAISS (vector search), Redis

Cloud & DevOps: AWS (EC2, RDS, S3, ALB, SageMaker, Bedrock, Auto Scaling), Docker, Git, CI/CD, GitHub Actions, SLURM

Tools & Platforms: ServiceNow (Skill Kit, GlideRecord, Update Sets), Flask, n8n, Weights & Biases, Jira, Pandas, NumPy

CS Fundamentals: Distributed Systems, System Design, Data Structures & Algorithms, OS Concepts, Networking, Concurrency

RESEARCH EXPERIENCE

Research Assistant, NLP Lab — *Feb 2026 – Present*
Texas A&M University — *College Station, TX*

- Built an automated data-ingestion and feature-engineering pipeline in Python converting raw cluster telemetry from 941 profiled jobs across 9 GPU configurations into a structured 37-feature dataset used for downstream model training.
- Extended the benchmark with a 1,884-run LLM inference dataset capturing prefill/decode split timing, TTFT, TPOT, and KV-cache pressure across 22 models and 4 GPU types.
- Trained gradient-boosted and neural regressors on the resulting dataset, reaching 23.7% MAPE ($R^2 = 0.881$) on training-job runtime and 8.8% MAPE ($R^2 = 0.982$) on inference latency.
- Instrumented distributed training runs across DistributedDataParallel, DataParallel, and DeepSpeed ZeRO stages 1–3 to capture multi-GPU scaling behaviour under mixed precision.

Graduate Researcher, NLP Lab — *Sep 2025 – Feb 2026*
Texas A&M University — *College Station, TX*

- Designed an 8-stage automated data processing pipeline converting engineering schematics to structured JSON graph outputs; integrated YOLOv8 for component detection, PaddleOCR with EasyOCR for text extraction, and HDBSCAN with UMAP for clustering.
- Containerized the pipeline as a REST API service with structured JSON output contracts, enabling integration with downstream reasoning and knowledge graph construction systems.

ACHIEVEMENTS & PUBLICATIONS

- **Hackathon Winner** – Best Idea Award, Shipsy Hackathon 2025 (40+ teams) for an autonomous standup-to-Jira pipeline; National Finalist, Smart India Hackathon competing against 500K+ participants across institute, regional, and national rounds.
- **AI/ML Club Sub-Lead**, IIIT Kottayam (2024) – Directed 8 ML workshops and 3 hackathons; grew active membership by 40% and mentored 30+ students. **IEEE Event Organizer** (2023) – Hosted the IoT Cloud Research Symposium with 200+ attendees across 6 research sessions.
- **Findings of EMNLP 2026 (Accepted)** – *Deep Learning Job Runtime Prediction for GPU Cluster Scheduling*; co-authored survey on deep learning job runtime prediction for GPU cluster scheduling, in collaboration with XPerf and Texas A&M University, grounded in a 941-job, 37-feature benchmark across 9 GPU configurations plus a 1,884-run inference dataset.
- **CBSE Merit Certificate** – Perfect score 100/100 in Mathematics (Grade 10), awarded to the Top 0.1% of all national candidates.